🎓 Professional Diploma in Node.js with JavaScript & TypeScript

Duration: ~42+ Hours (15 Sections, 229+ Lectures, Projects)
Format: Instructor-led + Hands-on Projects + Capstone
Skill Level: Beginner → Intermediate → Advanced → Production-ready

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Section 1: Getting Started with Node.js

- Introduction & Learning Path
- Installation & Setup (Node.js, npm, TypeScript, VS Code)
- Running JavaScript Outside the Browser
- Core Modules (fs, path, http, os)
- File System Operations (Sync & Async)
- Blocking vs Non-blocking I/O
- Creating a Web Server (JS + TS)
- Routing & Basic APIs
- Introduction to NPM & Package Management
- Code Quality Tools: ESLint + Prettier

Hands-on Project: *Basic HTTP Server with API in JS & TS*

Section 2: Web & Network Fundamentals

- Internet Basics: Client-Server Communication

- HTTP Protocol in Action
- Request/Response Cycle
- Frontend vs Backend Development
- Static vs Dynamic Content
- APIs & REST Concepts

Section 3: Node.js Core & Internals

- Node.js Architecture (V8, Libuv, C++)
- Processes, Threads & Thread Pool
- The Event Loop: Theory & Practical Demo
- Events & EventEmitter
- Streams: Readable, Writable, Duplex, Transform
- Buffering & Backpressure
- How `require` & `import` work (JS vs TS)

Hands-on Project: *Streaming File Server*

Section 4: Asynchronous Programming in Node.js

- Callback Functions & Callback Hell
- Promises (Creating & Consuming)
- Async/Await in Depth

- Error Handling in Async Code
- Running Multiple Promises (all, race, allSettled)

Hands-on Project: *File Processing Pipeline with Promises & Async/Await*

Section 5: Express.js Fundamentals

- Introduction to Express
- Express App Setup (JS + TS)
- REST API Design Principles
- CRUD Operations (GET, POST, PATCH, DELETE)
- Routing & Refactoring
- Middleware: Built-in, Custom & 3rd-Party
- Param Middleware & Chaining
- Serving Static Files
- Environment Variables & Config Management

Mini Project: *User Management API (JS & TS)*

Section 6: Databases with MongoDB & Mongoose

- MongoDB Overview (Local & Atlas Cloud)
- CRUD Operations with MongoDB Compass
- Connecting Node.js & Express to MongoDB
- Introduction to Mongoose

- Defining Models & Schemas
- CRUD with Mongoose Models

Mini Project: *Tours API with MongoDB & Mongoose*

Section 7: Advanced API Development

- MVC Architecture in Node.js
- Filtering, Sorting, Field Limiting & Pagination
- Advanced Query Features
- Aggregation Pipelines (Match, Group, Project, Unwind)
- Virtual Properties & Middleware (Pre/Post Hooks)
- Data Validation (Built-in & Custom Validators)

Project: *Advanced Tours API with Aggregation & Query Features*

Section 8: Error Handling & Debugging

- Debugging Node.js Apps with ndb
- Handling 404 & Invalid Routes
- Global Error Handling Middleware
- Handling Async Errors with Utility Wrappers
- Development vs Production Error Handling
- Database Error Handling (Duplicate Fields, Validation)
- Handling Unhandled Rejections & Uncaught Exceptions

Section 9: Authentication & Security

- User Authentication & Password Hashing (bcrypt)
- JWT Authentication & Session Management
- Protecting Routes with Middleware
- Authorization with User Roles & Permissions
- Password Reset Functionality (Token-based)
- Emailing with Nodemailer & SendGrid
- Security Best Practices:
    - Rate Limiting
    - Secure HTTP Headers (helmet)
    - Data Sanitization
    - Preventing Parameter Pollution
    - Storing JWT in Cookies (XSS & CSRF considerations)

Project: *Authentication System with JWT & Email-based Password Reset*

Section 10: Advanced Data Modeling

- MongoDB Data Modeling Techniques
- Geospatial Data & Queries (Maps & Locations)
- Embedding vs Referencing Documents
- Populating Related Data

- Virtual Populating (Tours & Reviews)
- Nested Routes in Express
- Factory Functions for Reusable Handlers
- Performance with Indexes
- Aggregations for Analytics & Stats
- Preventing Duplicate Entries

Project: *Tours + Reviews API with Geospatial Queries & Indexing*

Section 11: Server-Side Rendering with Pug

- SSR vs CSR (Server vs Client Rendering)
- Pug Template Engine Basics
- Layouts, Blocks, Partials
- Building Tour Overview & Details Pages
- Integrating Mapbox for Location Visualization
- User Authentication with Views
- Rendering Dynamic Error Pages
- User Account Page & Profile Updates

Section 12: File Uploads, Emails & Payments

- Handling File Uploads with Multer

- Resizing & Processing Images with Sharp
- Multiple File Uploads (Tours & Users)
- Advanced Email Templates with Pug
- Sending Real Emails via SendGrid
- Payment Integration with Stripe
- Handling Checkout Success & Webhooks
- Booking Model & API Development

Project: *Booking System with Image Uploads, Payments & Emails*

Section 13: Deployment & DevOps

- Git & GitHub Fundamentals
- Preparing a Node.js App for Production
- Deploying to Heroku, Render or AWS
- Enabling HTTPS & SSL
- Handling Process Signals (SIGTERM, SIGINT)
- Implementing CORS in Express
- Logging & Monitoring
- CI/CD Introduction (GitHub Actions, Jenkins)

Section 14: TypeScript for Node.js

- Setting up TypeScript in a Node Project
- tsconfig & Compiler Options
- Using Types with Express & MongoDB
- Creating Type-Safe Models & DTOs
- Error Handling with TypeScript
- Migrating JS Express App → TypeScript

Mini Project: *TypeScript-based REST API with Express + MongoDB*

Section 15: Advanced Topics & Capstone Project

- Testing Node.js Applications:
    - Unit Tests (Mocha, Jest)
    - Integration Tests (Supertest)
- WebSockets & Real-time Apps (Socket.IO)
- GraphQL with Node.js
- Caching Strategies (Redis Integration)
- Introduction to Microservices & Message Queues (RabbitMQ, Kafka)
- Dockerizing Node.js Apps
- Performance Optimization (Cluster, Load Balancing)
- Capstone Project:
    - Full Production-grade App (*Natours Booking App*)
    - Includes:

        - Authentication + Authorization
        - Tours + Reviews + Bookings APIs
        - Payments with Stripe
        - Email Notifications
        - File Uploads & Image Processing
        - Geospatial Queries + Maps
        - Secure Deployment

✅ Learning Outcomes

By the end of this program, learners will:

- Understand **Node.js internals** and its event-driven architecture.
- Build **scalable REST APIs** in both **JavaScript & TypeScript**.
- Integrate **MongoDB/Mongoose** with advanced data modeling.
- Implement **authentication, authorization & security best practices**.
- Build **real-time, payment-enabled, production-ready applications**.
- Deploy and maintain apps with **DevOps, CI/CD & Docker**